RESOURCE DETECTIVE: SOLVE THE ENERGY SOURCE CASE!

Squad Scorecard & Clue Tracker · Unit 2 Review Game

Detective Squad Name: _____ **Detectives:** _____

CATEGORY	$100	$200	$300	$400	$500	CLUE UNLOCKED
File 1 Resource Detective	☐	☐	☐	☐	☐	Clue #1: ___
File 2 Fossil Fuels	☐	☐	☐	☐	☐	Clue #2: ___
File 3 Turbines & Generators	☐	☐	☐	☐	☐	Clue #3: ___
File 4 Renewable Power	☐	☐	☐	☐	☐	Clue #4: ___
File 5 Resource Challenge	☐	☐	☐	☐	☐	Clue #5: ___

$300 = Team Challenge · $500 = Team Explain · Each correct clue earns the points AND unlocks that file's clue.

THE MYSTERY TOWN — CLUES

Complete a file to earn its clue. Use the clues to win the Final Challenge.

Clue 1	Clue 2	Clue 3	Clue 4	Clue 5
_____	_____	_____	_____	_____

TOTAL SCORE: _____ pts

TEACHER CARDS · Unit 2 Review Game

Cut these out. Team Explain ($500) = short spoken answer · Team Challenge ($300) = 30-second demo.

TEAM EXPLAIN CARDS ($500 CLUES)

Teams discuss, then explain in one or two spoken sentences. No long writing.

File 1 · Resource Detective — TEAM EXPLAIN

Why does it matter how quickly a resource is restored?

Expected: *A resource is renewable only if it comes back as fast as (or faster than) we use it.*

File 2 · Fossil Fuels — TEAM EXPLAIN

How are coal and petroleum alike? How are they different?

Expected: *Alike: both form from the remains of organisms over millions of years. Different: coal is solid from swamp plants; petroleum is liquid from ocean organisms.*

File 3 · Turbines & Generators — TEAM EXPLAIN

What stays the same in a coal plant and a nuclear plant?

Expected: *Heat boils water into steam → steam spins a turbine → the turbine turns a generator → electricity.*

File 4 · Renewable Power — TEAM EXPLAIN

Why is solar power renewable but not always available?

Expected: *The sun keeps shining (renewable), but solar depends on location, weather, and time of day (cloudy days, nighttime).*

File 5 · Resource Challenge — TEAM EXPLAIN

Name ONE renewable resource. Give one benefit AND one limitation, and where it works well.

Expected: *Example: wind — benefit: no fuel to burn; limitation: needs a windy location; works well where wind blows steadily.*

TEAM CHALLENGE CARDS ($300 clues)

30-second team demos. Act it out, then explain in one sentence.

File 1 · Resource Detective — TEAM CHALLENGE

Sort It Fast — call out a resource (coal, wind, sunlight, petroleum). The team points to RENEWABLE or NONRENEWABLE and says why.

Expected: *Renewable comes back quickly; nonrenewable does not.*

File 2 · Fossil Fuels — TEAM CHALLENGE

Three States — act out the three fossil fuels: solid (coal), liquid (petroleum), and gas (natural gas).

Expected: *Coal is solid, petroleum is liquid, natural gas is a gas.*

File 3 · Turbines & Generators — TEAM CHALLENGE

Spin the Path — spin your arms like a turbine, turn an imaginary generator, and shout "Electricity!"

Expected: *A spinning turbine turns the generator to make electricity.*

File 4 · Renewable Power — TEAM CHALLENGE

Four Sources — act out wind (spin arms), moving water (flow arms), sunlight (spread rays), and Earth's heat (rise from the ground).

Expected: *Wind, moving water, sunlight, and Earth's heat are all renewable.*

File 5 · Resource Challenge — TEAM CHALLENGE

Pick the Best — for a town with steady wind, cloudy days, and no river, act out wind power and explain your choice.

Expected: *Steady wind makes wind power the most reliable choice there.*